

Linear Algebra Bootcamp 3

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Problem 1. Let V be a vector space, and v a vector. Note $+v$ gives an invertible map $+v : V \rightarrow V$ by $w \rightarrow w + v$. What is the inverse?

Show this is not linear unless $v = 0$. Instead it is what is called "affine" show that if $a + b = 1$, $a, b \in F$, then $(+v)(aX + bY) = a(+v)X + b(+v)Y$, for X, Y in V .

Solution. The inverse of $+v$ is $-v : V \rightarrow V$ by $w \rightarrow w - v$. For all $w \in V$, $(+v)(-v)(w) = (+v)(w - v) = w - v + v = w$ and $(-v)(+v)(w) = (-v)(w + v) = w + v - v = w$.

Claim 1a. $+v$ is linear iff $v = 0$.

Proof. ($\implies$): Suppose $+v$ is linear. By the definition of linearity, for all $w \in V$ and $c \in F$ $(+v)(cw) = c(+v)(w)$. $(+v)(cw) = cw + v$ and $c(+v)(w) = c(w + v) = cw + cv$. If $cw + v = cw + cv$ then $v = cv$. This is only true if $v = 0$.

($\impliedby$): Suppose $v = 0$. Then for all $x, y \in V$ and $c, d \in F$, $(+v)(cx + dy) = cx + dy + v = cx + dy$. $c(+v)(x) + d(+v)(y) = c(x + v) + d(y + v) = cx + cv + dy + dv = cx + dy$. Thus by definition $(+v)$ is linear. ■

Claim 1b. If $a + b = 1$, $a, b \in F$, iff $(+v)(aX + bY) = a(+v)X + b(+v)Y$, for X, Y in V

Proof. ($\implies$): Suppose $a + b = 1$, $a, b \in F$. Then for all $X, Y \in V$ $(+v)(aX + bY) = aX + bY + v$. Since $a + b = 1$, then $aX + bY + v(a + b) = aX + av + bY + bv = a(X + v) + b(Y + v) = a(+v)(X) + b(+v)(Y)$

($\impliedby$): Suppose for some $a, b \in F$ $(+v)(aX + bY) = a(+v)X + b(+v)Y$, for X, Y in V . $(+v)(aX + bY) = aX + bY + v = a(+v)X + b(+v)Y = a(X + v) + b(Y + v) = aX + av + bY + bv = aX + bY + v(a + b) = aX + bY + v$. Then $v(a + b) = v$. By definition, $1 \in F$ is the only multiplicative identity, so $v * 1 = v$, thus $a + b = 1$. ■

Problem 2. A coset for a subspace W of V is a subset of form $(+v)(W)$ for some v , ie the set of vectors $\{v + w | w \in W\} =: v + W$. Say $X \sim Y$ if $X - Y \in W$. Show this is an equivalence relation, and $[v] = v + W$. E.g. Take $V = \mathbb{R}^3$ and W a line through the origin. What are the cosets? Note $v + W$ is in bijection with W (by the map $+v$). Let $Q := V / \sim$, which we write as V / W . Show that there is a unique vector space structure on Q so that the quotient map $V \rightarrow V / W$ is linear.

Solution.

Claim 2a. $X \sim Y$ if $X - Y \in W$ is an equivalence relation.

Proof. 1. **Reflexive:** Since W is a subspace of V . By definition of subspaces, W is not empty and thus there exists a $w \in W$. Furthermore by definition of subspaces, for all w in W , $0w \in W$ thus $0 \in W$. $X - X = 0$, thus $X - X \in W$ and $X \sim X$.

2. **Symmetric:** For all $X, Y \in V$, if $X \sim Y$ then $X - Y \in W$. Since W is a subspace then $-(X - Y) \in W$ or $Y - X \in W$. Thus $Y \sim X$.

3. **Transitive:** For all $X, Y, Z \in V$, if $X \sim Y$ and $Y \sim Z$ then $X - Y \in W$ and $Y - Z \in W$. Since W is a subspace, any linear combination is also in W thus $(Y - Z) + (X - Y) \in W$ or $X - Z \in W$. Thus $X \sim Z$. ■

Claim 2b. $[v] = v + W$

Proof. By definition of equivalence classes, it means for all $w \in [v]$, $v \sim w$. By definition, $v \sim w$, iff $a = v - w \in W$. By the Symmetric property of equivalence relations, $w \sim v$ thus $-a = w - v \in W$. w can be rewritten as $w = v - a = v + (w - v)$ for $-a \in W$. Thus $w \in v + W$. Since w can be any element of $[v]$, and every $w \in v + W$, it means $[v] = v + W$. ■

Claim 2c. Take $V = \mathbb{R}^3$ and W a line through the origin. What are the cosets?

Solution. The cosets of W , are any translation of W , or the line through the origin, by vector $v \in V = \mathbb{R}^3$. ■

Claim 2d. Let $Q := V / \sim$, which we write as V/W . Show that there is a unique vector space structure on Q so that the quotient map $V \rightarrow V/W$ is linear.

Proof. Let $Q := V / \sim$, which we write as V/W .

1. **Addition:** For all $[v], [w] \in Q$, then $[v] + [w] = [v + w]$
2. **Scalar multiplication:** For all $[v] \in Q$ and $c \in F$. Then $c[v] = [cv]$

Now let's apply these definitions to the properties of vector spaces to see if they hold.

1. Vector Addition:

- (a) **Commutativity of Vector Addition:** for all $[v], [w] \in Q$, $[v] + [w] = [v + w] = [w + v]$, by commutativity of vector addition in V . $[w + v] = [w] + [v]$ by definition of Addition in Q . Thus $[v] + [w] = [w] + [v]$
- (b) **Associativity of Vector Addition:** for all $[x], [y], [z] \in Q$, $([x] + [y]) + [z] = [x + y] + [z] = [(x + y) + z] = [x + (y + z)]$ because of associativity of vectors addition in V . $[x + (y + z)] = [x] + [y + z] = [x] + ([y] + [z])$ by def. of addition in Q . Thus $([x] + [y]) + [z] = [x] + ([y] + [z])$.
- (c) **Existence of additive identity or 0:** Let $0 \in V$ be the additive identity or zero. For all $[v] \in Q$, $[v] + [0] = [v + 0] = [v]$ and $[0] + [v] = [0 + v] = [v]$.

2. Scalar multiplication:

- (a) **Distributivity over Addition:** for all $[v], [w] \in Q$ and for all $c \in F$, $c([v] + [w]) = c[v + w]$ by the definition of addition in Q . $c[v + w] = [c(v + w)]$ by the def. of Scalar multiplication in Q . $[c(v + w)] = [cv + cw]$ by Distributivity over addition in V . $[cv + cw] = [cv] + [cw]$, by the definition of addition in Q . $[cv] + [cw] = c[v] + c[w]$ by the definition of Scalar multiplication in Q . Thus $c([v] + [w]) = c[v] + c[w]$.

- (b) **Distributivity over Scalar addition:** For all $[v] \in Q$ and for all $a, b \in F$, $(a + b)[v] = [(a + b)v]$ by definition of Scalar multiplication in Q . $[(a + b)v] = [av + bv]$ by the Distributivity of Scalar multiplication in V . $[av + bv] = [av] + [bv]$ by the definition of addition in Q . $[av] + [bv] = a[v] + b[v]$ by the definition of Scalar multiplication in Q . Thus $(a + b)[v] = a[v] + b[v]$.
- (c) **Compatibility with Scalar multiplication:** For all $[v] \in Q$ and for all $a, b \in F$. $(a * (b[v])) = a[bv]$ by the def. of Scalar multiplication in Q . $a[bv] = [abv]$ by the def. of Scalar multiplication in Q . $ab[v]$ by the def. of Scalar multiplication in Q . $ab[v] = (a * b)[v]$. Thus $(a * (b[v])) = ab[v]$.
- (d) **Scalar multiplication with identity:** For all $[v] \in Q$ and for $1 \in F$ s.t 1 is the multiplicative identity. $1[v] = [1v] = [v]$, by the definition of Scalar multiplication in Q and by the property of Scalar multiplication with identity in V .

Thus $Q := V/W$ is a vector space by definition. Let $f : V \rightarrow V/W$ be defined as follows. For all $v \in V$, $f(v) = [v]$. For all $v, w \in V$ and $a, b \in F$, $f(av + bw) = [av + bw] = [av] + [bw]$ by the definition addition in Q . $[av] + [bw] = a[v] + b[w]$ by the definition of Scalar multiplication in Q . $a[v] + b[w] = af(v) + bf(w)$. Thus $f(av + bw) = af(v) + bf(w)$. Thus the quotient map is linear. ■

Problem 3. Let $f : V \rightarrow Z$ be an F -linear map, with kernel K . Show that f factors uniquely through V/K , that the induced map $F : V/K \rightarrow Z$ satisfied $F(v + W) = f(v)$, and that this induced map is injective. Conclude f is surjective iff F is an isomorphism. Assuming V is finite dimensional, show $\dim V = \dim K + \dim V/K$.

Let $f : V \rightarrow Z$ be an F -linear map, with kernel K . By definition $V/K = V/\sim$ where $\sim$ is a equivalence s.t for all $\forall v, w \in V$ $v \sim w$ iff $v - w \in K$. Let $p : V \rightarrow V/K$ be the natural quotient map. By the above problem for all $[a] \in V/K$, $[a] = a + K$.

Claim 3a. f factors through p

Proof. For all $a, b \in V$, if $p(a) = p(b)$, by the definition of p , $[a] = [b]$. Thus by the definition of equivalence class, $a \sim b$. By the definition of the equivalence relation, $a - b \in K$. By the definition of a kernel, $f(a - b) = 0$. By the definition of linear maps, $f(a - b) = f(a) - f(b) = 0$. Adding $f(b)$ to both sides, we see $f(a) = f(b)$. Thus $p(a) = p(b) \implies f(a) = f(b)$. By Prehomework problem 4, f factors through p and there exists a $F : V/K \rightarrow Z$ s.t $f = F \circ p$. □

Henceforth assume F exists s.t $f = F \circ p$.

Claim 3b. f factors uniquely through p .

Proof. Assume the contrary, there exists a element $[a] \in Z/K$, s.t there exists no $v \in V$ s.t $p(v) = [a]$. By definition, a is a equivalence class where for every element $v \in [a]$ $v \sim a$ by the definition of equivalence classes. By the reflexive properties of equivalence relations, for every $v \in V$, $v \sim v$. Thus for every $v \in V$, $v \in [v]$. Thus $p(v) = [v]$ for every $v \in V$. Thus every element in Q , has at least 1 value which maps to it, thus forming a contradiction. Thus there does not exist a $[a] \in Z/K$ s.t there exists no $v \in V$ s.t $p(v) = [a]$. Thus p is surjective. By prehomework problem 4, since p is surjective then f factors uniquely through p . □

Henceforth assume F is unique and p is surjective.

Claim 3c. F is injective

Proof. Suppose for all $[a], [b] \in V/K$, that $F([a]) = F([b])$. Since $p(a) = [a]$ and $p(b) = [b]$. Then $F(p(a)) = F(p(b))$. By the definition of F , $f(a) = f(b)$. Thus $f(a) - f(b) = 0$. By linearity, $f(a - b) = 0$. By the definition of kernel, $a - b \in K$. By the definition of the equivalence relation, $a \sim b$. By the definition of equivalence classes, $b \in [a]$ and $a \in [b]$. By the equivalence relation homework problem question 2, equivalence classes are either pairwise disjoint or equal. Then since $b \in [a]$ and $a \in [b]$, $[a] = [b]$. Thus F must be injective by definition. $\square$

Henceforth assume F is injective.

Claim 3d. f is surjective iff F is an isomorphism.

Proof. ($\implies$): Suppose f is surjective. By definition it means, $F \circ p$ must be surjective. By the properties of composition of functions, since $F \circ p$ is surjective, F must be surjective as well. Since F is surjective and injective, by definition of bijective functions, F must be a bijection. By definition of isomorphism, since F is a linear map and is a bijection, it is also an isomorphism.

($\impliedby$): Suppose F was an isomorphism. Then by definition of isomorphisms, F must be a bijective linear map. By the definition of bijection, F must be surjective and injective. Since F is surjective, $F \circ g$ must also be surjective by the properties of composition of functions. By definition f is surjective $\square$

Henceforth assume V is finite dimensional

Claim 3e. Show $\dim V = \dim K + \dim V/K$.

Proof. By Problem 2 of LA 2, we know $\dim V = \dim K + \dim f_*(V)$. Since $f = F \circ p$, then $f_*(V) = F_*(p_*(V))$. Since p is surjective, $p_*(V) = V/K$. Thus $f_*(V) = F_*(V/K)$.

Since F is injective, by linear algebra bootcamp problem 5, $\ker F = 0$ and the dimension of the kernel of F is 0.

Let $m = \dim V/K$. By definition of dimension, there are m basis vectors in $\dim V/K$. Let $\{[v_1], \dots, [v_m]\}$ be the basis vectors in V/K .

Claim 3e.1. The set $\{F([v_1]), \dots, F([v_m])\}$ is linearly independent

Proof. Assume the contrary, the set $A = \{F([v_1]), \dots, F([v_m])\}$ is not linearly independent. By definition of linear independence, since A is not linearly independent, there exists a linear combination $a_1 F([v_1]) + \dots + a_m F([v_m]) = 0$ where there exists a $a_i \neq 0$ s.t $i \in [1, \dots, m]$. By linearity, $a_1 F([v_1]) + \dots + a_m F([v_m]) = F(a_1 [v_1] + \dots + a_m [v_m]) = 0$. Let $w = a_1 [v_1] + \dots + a_m [v_m]$, since at least 1 $a_i \neq 0$, $[v_i] \neq 0$ because if $[v_i] = 0$, then there exists a linear combination with non zero coefficients equal to 0. Thus $\{[v_1], \dots, [v_m]\}$ is linearly dependent. This is a contradiction, due to the fact that $\{[v_1], \dots, [v_m]\}$ is a basis and thus by definition linearly independent. By contradiction, $[v_i] \neq 0$. Thus $w \neq 0$. Since $F(w) = 0$, then $w \in \ker F$. Thus by contradiction, the set $\{F([v_1]), \dots, F([v_m])\}$ is linearly independent. $\square$

Claim 3e.1. The set $\{F([v_1]), \dots, F([v_m])\}$ spans $F_*(V/K)$.

Proof. For all $w \in V/K$. By definition of a basis, $\{[v_1], \dots, [v_m]\}$ spans V/K . Thus w can be represented as a linear combination of the elements in $\{[v_1], \dots, [v_m]\}$. So $w = a_1 [v_1] + \dots + a_m [v_m]$ where $a_i \in F$ for all $i \in [1, \dots, m]$. $F(w) = F(a_1 [v_1] + \dots + a_m [v_m])$. By the definition of linear map, $F(a_1 [v_1] + \dots + a_m [v_m]) = a_1 F([v_1]) + \dots + a_m F([v_m])$. Thus $\{F([v_1]), \dots, F([v_m])\}$ spans $F_*(V/K)$. $\square$

By the definition of basis, $\{F([v_1]), \dots, F([v_m])\}$ is linearly independent and spans $F_*(V/K)$, thus $\{F([v_1]), \dots, F([v_m])\}$ is a basis. Thus the $\dim F_*(V/K) = m$. Thus $\dim F_*(V/K) = \dim V/K$.

Thus $\dim V = \dim K + \dim V/K$. $\square$

Problem 4. Consider a system of linear equations, which we can view as the matrix equation $Ax = b$ (for some $n \times m$ A , and $b \in F^n$, here X is the "variable"). Show that the solution set is either empty, or a coset for the kernel – hopefully this reminds you of something from linear algebra.

Proof. Let $A \in M_{n \times m}$ s.t

$$A = \begin{bmatrix} c_1 & \cdots & c_m \end{bmatrix} \quad (1)$$

Where c_i for $i \in [1, \dots, m]$ is the i th column of A . Let $x \in F^m$ and $b \in F^n$. b can either be in or not in the column space of A .

1. If $b \notin \text{col}A$: Then there exists no linear combination of $c_1, \dots, c_m$ equal to b . Thus $Ax = \sum_{i=1}^m x_i c_i \neq b$ for any $x \in F^m$. Thus there exists no $x \in F^m$, s.t $Ax = b$. Thus the solution set of $Ax = b$ is empty.
2. If $b \in \text{col}A$: Then there exists a linear combination of $c_1, \dots, c_m$ equal to b . Thus there exists a $x \in F^m$, s.t $Ax = \sum_{i=1}^m x_i c_i = b$. Thus the solution set is not empty. Let v be an element in the solution set of $Ax = b$, so $Av = b$. There always exists a solution to $Ax = 0$, $x = 0$ so $Ax = \sum_{i=1}^m 0c_i = 0$. Thus the $\ker A \neq \emptyset$. Let $\{n_1, \dots, n_k\}$ be the basis elements of $\ker A$. By definition of a basis, $\{n_1, \dots, n_k\}$ spans $\ker A$. For every $w \in \ker A$, w can be represented as a linear combination of the elements of $\{n_1, \dots, n_k\}$ so $w = a_1 n_1 + \dots + a_k n_k$ for $a_i \in F$. Furthermore by definition of kernel, since $w \in \ker A$, $Aw = 0$. Thus for all $w \in \ker A$, $A(v + w) = Av + Aw$ by linearity. Since $Av = b$ and $Aw = 0$, $A(v + w) = b + 0 = b$. Thus for all $w \in \ker A$, $v + w$ is a solution to $Ax = b$. By the definition of coset, $v + \ker A = \{v + w \mid \forall w \in \ker A\}$. Thus by definition, all the elements of $v + \ker A$ are solutions to $Ax = b$.

Claim 4a. There exists no $u \notin v + \ker A$ s.t $Au = b$.

Proof. Let $V := F^m$, $Z := \text{col}A$, and $Q := F^m / \ker A = F^m / \sim$, where $\sim$ is an equivalence relation defined as for all $a, b \in F^m$ $a \sim b$ iff $a - b \in \ker A$. Let $f : F^m \rightarrow \text{col}A$, be the linear map associated with the matrix A . Let $p : F^m \rightarrow Q$ be the natural quotient map, s.t $p(x) = [x] = x + \ker A$ for all $x \in F^m$. Finally let $F : Q \rightarrow Z$ be a linear map s.t $f = F \circ p$, thus f factors through p . By the previous problem we proved that F is injective and unique. Assume the contrary, $f(w) = Aw = f(u) = Au = b$ s.t $w \notin v + \ker A$ but $u \in v + \ker A$. Since $f = F \circ p$, $f(w) = F(p(w)) = F(w + \ker A)$ and $f(u) = F(p(u)) = F(u + \ker A)$. Since $u \in v + \ker A$, then $u + \ker A = v + \ker A$. Thus $F(u + \ker A) = F(v + \ker A)$. Thus $F(w + \ker A) = F(v + \ker A)$. Since F is injective, by definition, $w + \ker A = v + \ker A$. Thus a contradiction forms since $w + \ker A = v + \ker A$ and $w \notin v + \ker A$. Thus by contradiction, for all w s.t $Aw = b$, $w \in v + \ker A$. $\square$

Thus the solution set is $v + \ker A$, or a coset of the kernel. $\square$

Problem 5. Let $f : V \rightarrow W$ be a surjective map of sets, and $\sim$ the corresponding equivalence relation. Show that $f_* : \text{Pow}(V) \rightarrow \text{Pow}(W)$ and $f^* : \text{Pow}(W) \rightarrow \text{Pow}(V)$ induce inverse bijections between $\text{Pow}(W)$ and the subset of $\text{Pow}(V)$ consisting of subsets "closed under $\sim$ ", meaning subsets S of V such that if $s \in S$, and $x \sim s$ then $x \in S$. Now suppose f is a surjective linear map between vector spaces. Show these same maps give a bijection between subspaces of W and subspaces of V that contain the kernel. Let K be a subspace of V . Explain why the set of V/K are in natural bijection with subspaces of V containing K .

Claim 5a. $f_* : \text{Pow}(V) \rightarrow \text{Pow}(W)$ and $f^* : \text{Pow}(W) \rightarrow \text{Pow}(V)$ induce inverse bijections between $\text{Pow}(W)$ and the subset of $\text{Pow}(V)$ consisting of subsets "closed under $\sim$ ", meaning subsets S of V such that if $s \in S$, and $x \sim s$ then $x \in S$.

Proof. Let S be the set of subsets of V which are closed under " $\sim$ ". For all $X, Y \in S$, for all $a \in X$ and $b \in Y$. If $f(a) = f(b)$, then by definition of S , $a \sim b$, $b \in X$, and $a \in Y$. Define a function $g : S \rightarrow \text{Pow}(W)$ as the following: for all $X \in S$

$$g(X) = f_*(X) \quad (2)$$

If for all $X, Y \in S$, $g(X) = g(Y)$ then $f_*(X) = f_*(Y)$ by definition. Then by definition of images, for all $x \in X$ there exists a $y \in Y$ s.t. $f(x) = f(y)$ thus $x \sim y$ thus $x, y \in X$ and $x, y \in Y$ since the sets of S are closed under the equivalence relation. Similarly, for all $y \in Y$ there exists a $x \in X$ s.t. $f(x) = f(y)$ thus $x \sim y$ thus $x, y \in X$ and $x, y \in Y$ since the sets of S are closed under the equivalence relation. Since all the elements in X and Y are in both X and Y , $X = Y$. Thus g is injective.

Define $h : \text{Pow}(W) \rightarrow S$ as the following. For all $U \in \text{Pow}(W)$:

$$h(U) = f^*(U) \quad (3)$$

For all $X \in S$, $(h \circ g)(X) = h(g(X)) = h(f_*(X))$, by definition of g . Assume the contrary, $f^*(f_*(X)) \neq X$, then there exists an element $a \in f^*(f_*(X))$ s.t. $a \notin X$. By definition of a preimage, it means $f(a) \in f_*(X)$. Furthermore by definition of an image $f_*(X) = \{f(x) | x \in X\}$. Thus there exists a $x \in X$ s.t. $f(a) = f(x)$. By definition of the equivalence relation, $x \sim a$ and $a \in X$. This forms a contradiction, thus there exists no $a \in f^*(f_*(X))$ but not in X . Assume the contrary, exists a $a \in X$ s.t. $a \notin f^*(f_*(X))$. By definition of an image, $f(a) \in f_*(X)$. By definition of a preimage, for $T \in \text{Pow}(W)$ $f^*(T) = \{s \in V, f(s) \in T\}$. Since $f(a) \in f_*(X)$, by definition of a preimage, $a \in f^*(f_*(X))$. Thus forming a contradiction. By contradiction there exists no $a \in X$ but not in $f^*(f_*(X))$. Thus for all $x \in f^*(f_*(X))$, $x \in X$ and for all $x \in X$, $x \in f^*(f_*(X))$. Thus $f^*(f_*(X)) = X$ and $(h \circ g)(X) = X$.

For all $T \in \text{Pow}(W)$, $(g \circ h)(T) = g(h(T)) = g(f^*(T)) = f_*(f^*(T))$, by definition of h and g . Assume the contrary, there exists a $t \in T$ s.t. $t \notin f_*(f^*(T))$. Since f is surjective, there exists a $a \in V$ s.t. $f(a) = t$. By definition of a preimage, $a \in f^*(T)$. Thus by the definition of an image, $f(a) \in f_*(f^*(T))$ thus $t \in f_*(f^*(T))$. Thus a contradiction forms. By contradiction, there exists no $t \in T$ s.t. $t \notin f_*(f^*(T))$. Similarly, assume the contrary, there exists a $t \in f_*(f^*(T))$ s.t. $t \notin T$. By definition of image, there exists $a \in f^*(T)$ s.t. $f(a) = t$. Since $a \in f^*(T)$, and $f(a) = t$, by definition of a preimage $f(a) \in f_*(T)$ thus $t \in T$. Thus by contradiction there exists no $t \in f_*(f^*(T))$ s.t. $t \notin T$. Thus for all $t \in f_*(f^*(T))$, $t \in T$ and for all $t \in T$, $t \in f_*(f^*(T))$. Thus $f_*(f^*(T)) = T$ and $(g \circ h)(T) = T$.

Thus g and h are inverses. By equivalence relation problem 8, g and h are bijections. $\square$

Claim 5b. Now suppose f is a surjective linear map between vector spaces. Show these same maps give a bijection between subspaces of W and subspaces of V that contain the kernel.

Proof. 1. **f^* and Subspaces of W :** Let U be a subspace of W . Since f is a surjective linear map, every element in U has a preimage in V . For all $v_1, v_2 \in f^*(U)$, by definition of preimage, $f(v_1), f(v_2) \in U$. Then $f(v_1) + f(v_2) = f(v_1 + v_2) \in U$, because U is a subspace and closed under addition and by definition of a linear map. By definition of a preimage, $v_1 + v_2 \in f^*(U)$. Thus $f^*(U)$ is closed under addition. For all $v \in f^*(U)$ and for all $c \in F$. That means $f(v) \in U$. Since U is subspace, $c * f(v) \in U$. By definition of linear maps, $cf(v) = f(cv) \in U$. Thus by definition of a preimage, $cv \in f^*(U)$. Thus $f^*(U)$ is closed under scalar multiplication. Finally since f is a linear map, it preserves the 0 vector. Therefore $f(0_v) = 0_w$. Since $0_w \in U$, because W is subspace, $0_v \in f^*(U)$. Moreover for all $k \in \ker f$, $f(k) = 0_w \in U$ by definition. Thus $k \in f^*(U)$ for all $k \in \ker f$, thus $\ker f \subset f^*(U)$. Thus $f^*(U)$ is a subspace with the kernel of f .

2. **f_* and Subspaces of W :** Let X be a subspace of V with the kernel. For all $x, y \in X$, $f(x), f(y) \in f_*(X)$ by definition. Since X is a subspace and closed under linear combinations, for all $a, b \in F$, $ax + by \in X$, then $f(ax + by) \in f_*(X)$. By definition of linear maps, $f(ax + by) = af(x) + bf(y) \in f_*(X)$. Thus $f_*(X)$ is also closed under linear combination. Thus $f_*(X)$ is a subspace in W .

For all subspaces U in W , $f_*(f^*(U))$. By definition of preimage, $f^*(U)$ is the set of all $v \in V$, s.t $f(v) \in U$. This means $f^*(f_*(U))$ contains all $f(v)$ for $v \in f^*(U)$. But since $f(v) \in U$ for these v , $f_*(f^*(U))$ contains all elements of U . Thus $U \subset f_*(f^*(U))$. For all $u \in f_*(f^*(U))$, then $u = f(v)$ for some $v \in f^*(U)$. By definition of preimage and the fact that f is function and can only assign a element in V to 1 element in W , $f(v) = u \in U$. Thus $f^*(f_*(U)) \subset U$. Thus since $U \subset f_*(f^*(U))$ and $f^*(f_*(U)) \subset U$, $f_*(f^*(U)) = U$

For all subspace X in V which contain kernel f . For all $x \in X$, $f(x) \in f_*(X)$ by definition of a image. By definition of a preimage, since $f(x) \in f_*(X)$, $x \in f^*(f_*(X))$. Thus for all $x \in X$, $x \in f^*(f_*(X))$. Thus $X \subset f^*(f_*(X))$. For all $x \in f^*(f_*(X))$, by definition of a preimage $f(x) \in f_*(X)$. Let $y \in X$, s.t $f(y) = f(x)$. Since f is a linear map, represent its equivalent matrix as F . By the previous problem we know that the set of the solutions to $Fb = f(x)$ for $b \in V$ are in the coset of $b + \ker X$. Since $\ker f \in X$ and X is a subspace and is thus closed under addition, and $Fy = f(y) = f(x)$ is a valid solution all solutions to $Fb = f(x)$ must be in X . Thus since $Fx = f(x) = f(x)$ $x \in X$. Thus for all $x \in f^*(f_*(X))$, $x \in X$. Thus $X \subset f^*(f_*(X))$ and $f^*(f_*(X)) \subset X$. Thus $X = f^*(f_*(X))$.

Thus f_* and f^* induce bijections between subspaces of W and subspaces of V that contain the kernel. $\square$

Let K be a arbitrary subspace in V .

Claim 5c. V/K are in natural bijection with the subspaces containing K .

Solution. $p : V \rightarrow V/K$ is the natural quotient map.

Let A be the set of subspaces of V containing K . Then $g : A \rightarrow \text{Pow}(V/K)$ is defined as follows, for all $X \in A$, $g(X) = p_*(X)$. If for all $X, Y \in A$, $g(X) = g(Y)$, then for all $x \in X$, there exists a corresponding $y \in Y$ s.t $x + K = y + K$. By the definition of X/K as the set of equivalence classes under $\sim$ where if $a \sim b$ then $a - b \in K$. $x + K = [x] = [y] = y + K$. By definition of equivalence classes, $x \sim y$. Since this applies to each element of X , $X = Y$.

Let $h : \text{Pow}(V/K) \rightarrow A$ where for all $T \in \text{Pow}(V/K)$, $h(T) = p^*(T) = \bigcup_{v+K \in T} v + K$. In other words, $h(T)$ is the set of all vectors who are in the cosets within T .

For $X \in A$, $(h \circ g)(X) = h(g(X)) = h(p_*(X)) = p^*({x+K \mid x \in X}) = \bigcup_{x+K \in \{x+K \mid x \in X\}} x + K = X$. For $T \in \text{Pow}(V/K)$, $(g \circ h)(T) = g(h(T)) = g(\bigcup_{v+K \in T} v + K) = p_*(\bigcup_{v+K \in T} v + K) = T$.

Thus g and h are natural inverses. Thus there exists a natural bijection from $\text{Pow}(V/K)$ to A . $\blacksquare$